

A Smart Tutor Avatar for Mimicking Characters of Bangla Sign Language

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Abstract—A smart tutor avatar teaches Bangla Sign Language (BSL) by mimicking its 49 characters, providing a visual and immersive learning experience to bridge communication gaps for the hearing-impaired. This research introduces a novel avatar system designed specifically for BSL, enhancing the learning process through interactive and engaging methods. The avatar uses machine learning and image processing to analyze hand gestures and replicate BSL signs, creating an interactive learning tool. Initial dataset pre-processing trains the system to produce accurate sign visuals. The user interface is intuitive, allowing learners to engage in physical exercises to improve sensorimotor skills, with immediate feedback provided via cosine similarity. While avatar tutors have been explored for American Sign Language, this study is pioneering in its approach to BSL. The proposed research has the potential to make BSL learning more accessible and enjoyable for a wider audience.

Index Terms—Bangla Sign Language; Image processing; YOLOv8; Tutor Avatar; Cosine similarity

I. INTRODUCTION

One of the biggest components of human life is communication, it acts as a bridge for human communication and social interactions among people all around the world. Although communication plays a huge role in the social interaction and quality of life as a human being, there are different ways of language through interaction. One may talk verbally, others may use signs or written forms of language as a communication medium. A significant percentage of people in Bangladesh use sign language as their main medium of communication. For example, Bangladesh has 133,517 people with hearing impairments and 194,394 people with voice problems, according to the Department of Social Services [1]. It may seem to be a mere medium of communication to us but for many, it is like a bridge between them and the outside world. These communication techniques or how efficient it will

be to connect to the outside world solely depend on how proficient in sign language the hearing-impaired people and others accompanying them are.

To address this special need, the research introduces an effective and innovative technological idea: the "Smart Avatar Tutor for Mimicking Characters of Bangla Sign Language." An intelligent avatar that can mimic Bangla sign language can help to teach these people who are without knowledge of sign language. This will set a significant advancement in assistive online educational technology by incorporating the YOLOv8 Nano baseline model. YOLOv8 Nano baseline model is a simpler version of the "You Only Look Once" series and is vastly known for its effectiveness and precision in object identification and also the lightness of the model, especially in settings with confined processing capacity. This model is very suitable for identifying and analyzing the complex motions of Bangla Sign Language (BSL) due to its real-time object identification capabilities. Given the YOLOv8 Nano baseline model, the avatar is capable of mimicking these actions and will provide users with a more effective, accurate, and entertaining learning tool.

The distinguishing feature of this paper is the unique application of the YOLOv8 Nano Baseline model in the industry of sign language teaching technologies. although there have been other attempts to add technology in sign language teaching platforms mainly using CNNs, RNNs, 3D CNNs, Pose Estimation Models, Combining different types of neural networks, such as CNNs for feature extraction and RNNs/LSTMs for sequence modeling, and so on. However, it showed the highest accuracy using the Yolo model series. YOLOv8 performs efficiently in the case of real-time image detection for sign language. But we choose to use the lightweight model yolo8 nano baseline model for this new learning platform by considering the computational complexity, and storage complexity as a

concern.

In this article, Section II provides a brief literature review on fields related to Bangla Sign Language detection. The datasets used to train our model and the system design are described in Section III. Section IV demonstrates the implementation and analysis of the results. Finally, Section V concludes with the hope of real-world applications for the trained model, featuring a smart tutor avatar mimicking Bangla Sign Language.

II. LITERATURE REVIEW

Leveraging advanced machine learning algorithms, sign language detection has reached significant milestones. The use of avatars to teach sign language is an approach that has been adopted in numerous research projects in the past and is increasingly being utilized in the present. Quandt et al. [2] developed SAIL (Signing Avatar and Immersive Learning), a system built in Unity3D that used motion-capture-based signing avatars with more natural facial expressions to teach American Sign Language (ASL) in an immersive 3D environment. Rahaman et al. [3] used CNN and LSTM algorithms, combined with OpenPose to create a 3D avatar for demonstrating Bangla Sign Language (BSL) gestures, allowing users to select Bangla numbers or alphabets on their website, practice the corresponding signs, and receive feedback on their performance accuracy through cosine similarity to determine if they can progress to the next word, highlighting the effectiveness of machine learning in sign language recognition. Lee et al. [4] developed an ASL learning app prototype in the form of a whack-a-mole game with real-time sign recognition, utilizing an LSTM RNN combined with k-Nearest Neighbors (KNN) to handle static and dynamic datasets but it was limited to improving accuracy. Shafiqul et al. [5] developed a CNN-based model with a dataset of over 31,000 Bangla character and numerical samples, achieving high accuracy but with limitations in real-world generalization. This new research introduces a smart tutor avatar that mimics Bangla sign language, addressing these limitations by providing an interactive learning experience tailored to the needs of Bangla sign language users. Islam et al. [6] used CNNs to decode static Bangla Sign Language gestures, achieving 88% accuracy. Basnin et al. [9] combined CNN and LSTM models, showcasing the potential for improved sign language recognition by effectively addressing both spatial and temporal aspects of the data. In a study on traffic signal detection, YoloV5, YoloV7, and YoloV8 were tested to address the challenge of reducing accidents in the growing field of autonomous vehicles, with YoloV8 achieving 14% and 13% improvements in mAP50-95 compared to the other models, based on analysis of precision, recall, mAP, confusion matrix, and F1 score [7].

Researchers and scientists adopted several strategies for real-time object detection. They have obtained a good amount of accuracy for sign language detection in various languages. For Bangla Sign Language, a research work

has been conducted using YOLOv7 that achieved mAP@.5 accuracies in detecting various classes from 85 percent to 97 [11]. However, there are only a few researches in discovering a smart sign language tutor.

In the literature, the majority of research primarily emphasizes traditional deep learning techniques for sign language recognition. Only a limited number of studies have leveraged advanced pre-trained models, such as YOLO, which demonstrate improved accuracy in detecting complex sign language gestures. These models, particularly YOLOv5 and YOLOv8, are better suited for real-time applications due to their enhanced performance in object detection. However, the application of YOLO in real-time Bangla sign language detection remains underexplored, with few studies investigating the feasibility of integrating these models into smart avatar tutors for various sign languages. Specifically, there is a notable absence of research focused on implementing a trained YOLO model to create a smart avatar tutor that mimics Bangla Sign Language, presenting a promising opportunity for future research and development in this area.

III. METHODOLOGY

The proposed approach consists of a couple of steps. Firstly, prepare the collected datasets and assess the performance in advanced machine learning models. Secondly, design and implement a smart system that can teach bangla sign language. Lastly, integrate the model with the system. The primary goal was to train the model and ensure high accuracy so that in real time detection, the drop of accuracy is slight.

The figure 1 illustrates the work plan of this research. Initially, data was collected and preprocessed, with appropriate labeling. The data was then split into two segments with an 80:20 ratio—80% for training and 20% for testing. The YOLOv8 Nano baseline model was used for training. Finally, the model was integrated into a smart avatar system designed for teaching Bangla Sign Language.

A. Dataset

Obtaining a dataset for Bangla sign language was a challenging task. Popular sign languages such as ASL have more available datasets. After a lot of searching, we found a dataset that is publicly available from the website ScienceDirect [8]. The dataset comprises 29,428 images of hand gestures for Bangla numbers and alphabets, encompassing 49 classes of Bangla sign language characters, with an equal distribution of 14,745 images for both detection and recognition tasks.

B. Model Specification

YOLO (You Only Look Once) is an efficient object detection algorithm known for its speed. Unlike traditional methods that analyze images sequentially, YOLO applies a single convolutional neural network (CNN) to the entire image at once, dividing it into a grid where each cell

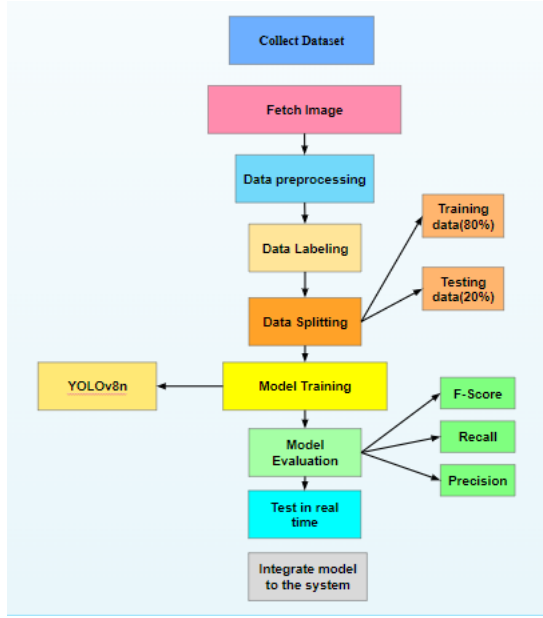


Figure 1: Top-Level Overview of the Proposed System

predicts bounding boxes and associated probabilities. This real-time detection approach is faster and more efficient. YOLOv8 performs very well in real-time object detection. Therefore, it is the preferred model. It utilizes modern innovations like CSP (Cross Stage Partial Networks) and PAN (Path Aggregation Network) but in a more streamlined form. [10]

A study compared YOLO V5, V7, and V8 for traffic sign detection using a dataset of 4650 images across 15 traffic signs. The results showed YOLOv8 outperformed the others, with a 14% and 13% improvement in mAP50-95 over YOLO V5 and V7, respectively [7]. In conclusion, the YOLOv8 performs relatively better than all previous YOLO variants.

C. Bangla Sign Language Avatar System Design

The activity diagram in Fig 2. depicts the system workflows and how user will interact with the system.

- **Training the Dataset with the ML Model:**

- The process initiates with training the dataset using a machine learning model.

- **Validation of Model Training:**

- Upon completion of the training process, the system verifies the success of the model training.
 - * **If successful:** The model is subsequently integrated into a Python-based system.
 - * **If unsuccessful:** The model is revised and retrained before further steps are taken.

- **Model Integration into Python System:**

- Following successful training, the model is integrated into the Python-based system for further utilization.

- **User Interface Implementation:**

- A user interface is implemented to facilitate interaction between the end-user and the system.

- **User Interaction with the System:**

- The user engages with the system via the implemented interface, enabling interaction with the trained model.

- **Performance Evaluation:**

- The system evaluates user performance during the interaction phase, providing real-time feedback and monitoring progress.

- **Assessment of Learning Effectiveness:**

- Based on performance evaluation, the system assesses whether the user is learning effectively.
 - * **If effective:** The system progresses to provide advanced lessons.
 - * **If ineffective:** The system prompts the user to revisit previous steps or offers additional opportunities for improvement.

- **Completion of Training:**

- The training process is considered complete once all necessary learning objectives have been achieved.

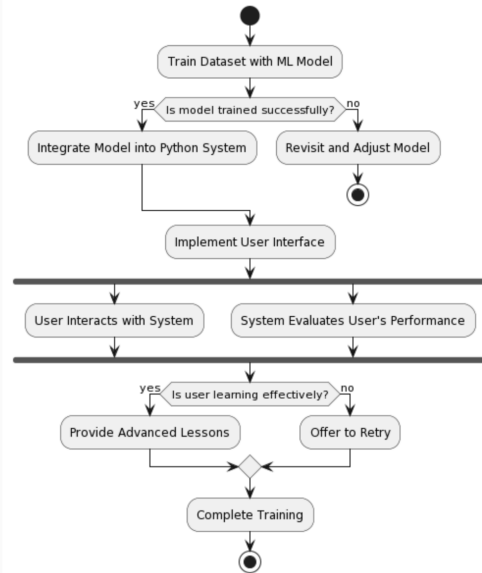


Figure 2: Activity Diagram of A Smart Tutor Avatar

D. Avatar Images

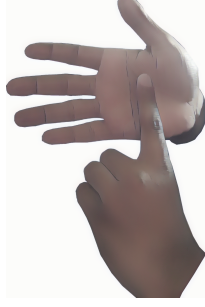
In figure 3, we processed the images into an avatar image. For the avatar hand gesture images, we captured pictures representing Bangla Sign Language gestures and evaluated them using the trained model. Initially, we removed the background of the images and necessary rotation for clearer visualization using Adobe Illustrator. We then transformed these images into 3D avatar representations, allowing users to mimic the gestures more accurately and effectively.



(a)



(b)



(c)

Figure 3: Avatar image: (a) BSL hand gesture image, (b) Background removed and rotated BSL image hand gesture sample, (c) 3D avatar converted BSL image

IV. IMPLEMENTATION AND RESULT ANALYSIS

A. Result Analysis

In this research, YOLOV8 nano variant model is used to train the image dataset for real-time object detection. YOLOv8 nano variant is the smallest of the YOLO V8 variants. This model is faster and runs efficiently in low-power device. This variant is very optimized for real-time inference with low latency that makes it ideal for the designed system. After training the dataset with YOLOV8 nano variant model, we got a notable result. The TABLE I. provides a few sample results of individual bangla sign language characters in validation dataset.

Fig 4 shows the loss function for the training dataset. At 20 epochs the loss curve gets stiffer. Similarly, in Fig 6 shows the validation loss function

Analysis of Metrics -: To assess the model's performance in detecting Bangla Sign Language (BSL) characters, we used two essential metrics: mAP50(B) and mAP50-95(B). The mAP50(B) (Mean Average Precision at 50% IoU) metric assesses the model's precision when the Intersection

Table I: Validation Results of Each Character

Label	Character	Box(Precision)	Recall	mAP50	mAP50-95
0	০	0.789	0.933	0.852	0.790
1	১	0.984	1	0.995	0.918
2	২	1	0.983	0.991	0.940
3	৩	0.879	0.967	0.961	0.909
4	৪	1	0.833	0.917	0.880
5	৫	0.968	1	0.994	0.902
6	৬	0.982	0.917	0.956	0.855
7	৭	1	0.983	0.991	0.843
8	৮	1	1	0.995	0.897
9	৯	0.981	0.883	0.940	0.909
10	১০	1	1	0.995	0.787

over Union (IoU) threshold is set to 50%, a common benchmark for object detection tasks. This statistic indicates how accurately this model can detect BSL characters with a relatively low overlap requirement. On the other hand, mAP50-95(B) (Mean Average Precision at IoU thresholds ranging from 50% to 95%) provides a more comprehensive evaluation of the model's performance across varying levels of overlap. By considering different IoU thresholds, this metric accounts for stricter detection requirements, allowing us to assess the robustness and generalization ability of the model under different scenarios. The mAP50(B) and mAP50-95(B) curves illustrate the model's learning progress across epochs. Smooth curves indicate stable convergence, while fluctuations reflect changes in generalization. These trends are essential for evaluating performance and guiding hyperparameter tuning and optimization. Figure 5 shows the model's metrics.

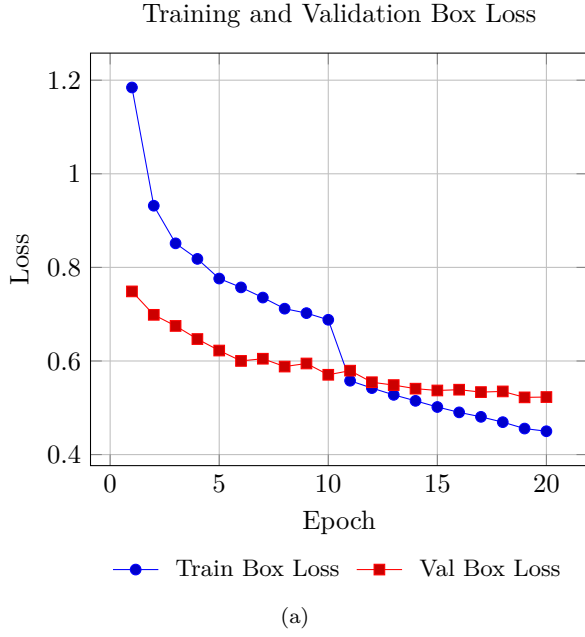
The performance metrics of an object detection model trained to recognize various Bangla characters and numerals. The evaluation was conducted over 2940 images, encompassing 2939 instances of different classes. The key performance indicators used are Precision (Box P), Recall (R), mean Average Precision at 50% IoU threshold (mAP50), and mean Average Precision across IoU thresholds from 50% to 95% (mAP50-95).

- **General Accuracy:** The model achieved high overall precision (0.981) and recall (0.951), indicating strong detection capabilities across all classes.
- **mAP50:** The mean Average Precision at 50% IoU (Intersection over Union) is 0.968, suggesting that the model is highly accurate in detecting objects with at least 50% overlap with ground truth.
- **mAP50-95:** The mean Average Precision across a range of IoU thresholds (0.87) demonstrates the model's robustness and accuracy in various conditions.

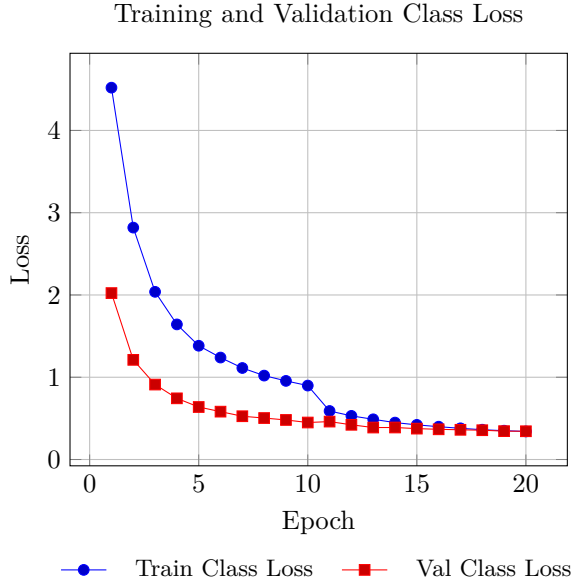
The following table shows the performance metrics of the object detection model validation on various classes:

B. Model Integration and System Implementation

After analyzing and selecting the perfect model for the smart avatar tutor teaching Bangla sign language, we exported the YOLOv8 Nano Baseline model in Py-Torch



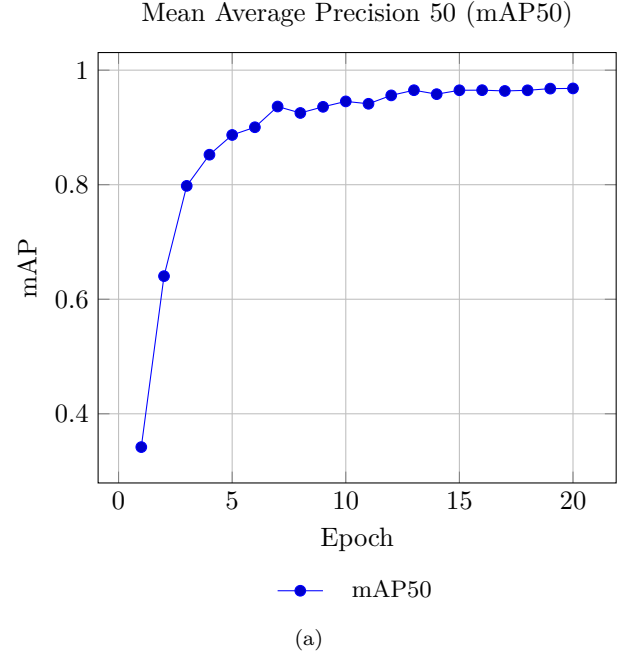
(a)



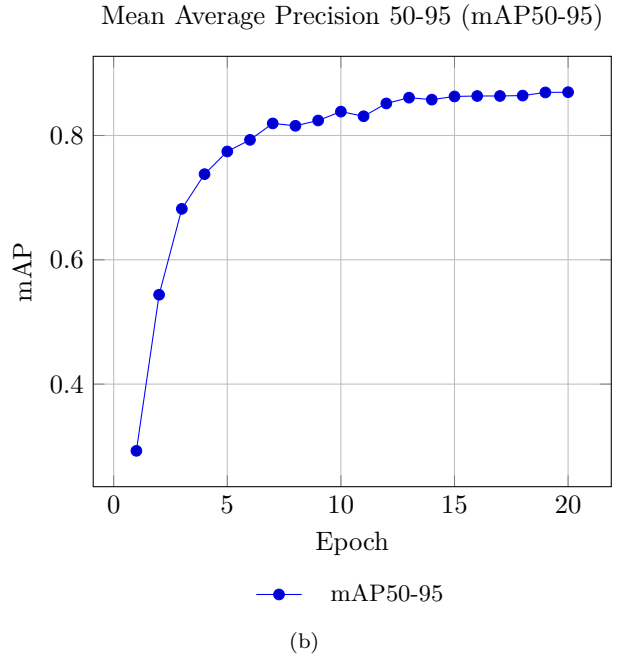
(b)

Figure 4: Training and Validation Losses: (a) Training and validation box loss over 20 epoch, (b) Training and validation class loss over 20 epoch

format. Then we converted the model into ONNX model format to integrate with this system. Later, we developed a Python desktop app using opencv-python, onnxruntime, and PyQt5 to develop the smart avatar tutor system. Onnxruntime utilizes CPU to load the model. The smart avatar tutor system initially displays a character sign language image with the character name followed by the accuracy on the right side of the screen. Then the user opens his camera to mimic the Bangla sign language



(a)



(b)

Figure 5: mAP Metrics over Epochs: (a) mAP50 over 20 epochs, (b) mAP50-95 over 20 epochs

character. Initially, the threshold is 70%. Due to the environment and lighting being different from the dataset we trained, the accuracy does not always give the optimal result. Therefore, there might be some discrepancy in the accuracy. However, the system was able to detect up to 95% in an environment where there is a proper amount of

lighting. So, the system is quite user-friendly to use and learn all 49 Bangla Sign Language characters.

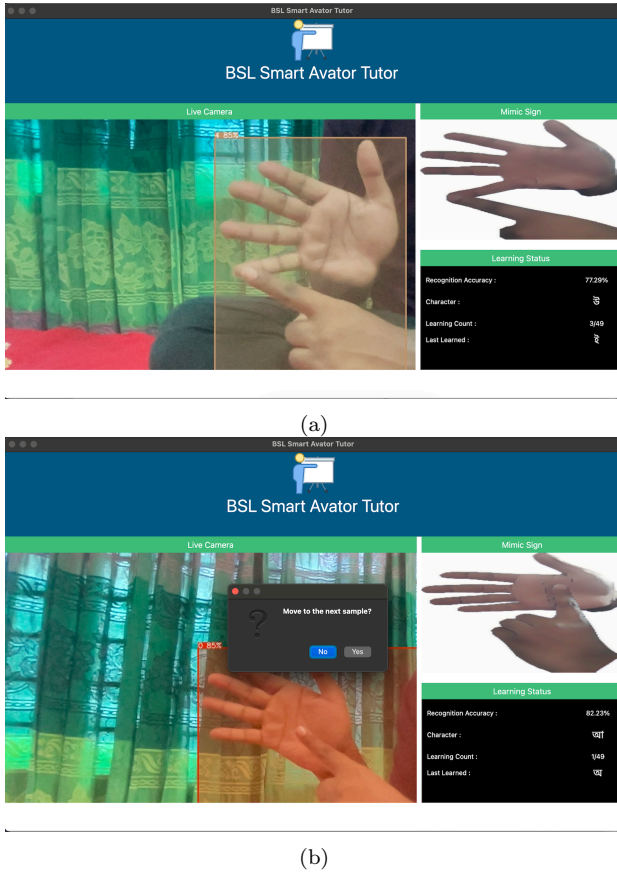


Figure 6: A Smart Avatar Tutor: (a) Recognizing the Bangla sign language character "উ" from a hand gesture, (b) User successfully learning the Bangla sign language character "আ"

Fig. 6 shows the live demonstration of the smart avatar tutor system where a user is learning sign language with various steps in a user-friendly user-interface. Fig. 6 (a) demonstrates an example of the system where it is trying to detect bangla sign language character "উ". Figure 6 (b) illustrates another instance of the smart avatar tutor, where it successfully recognized the Bangla sign language character "আ", and the smart avatar tutor is prompting the user to proceed to the next character.

V. CONCLUSION

This research aimed to create a teaching system for the hearing-impaired community in Bangladesh, focusing on Bangla Sign Language (BSL). Recognizing that BSL proficiency is crucial for effective communication, we integrated the Yolov8n model, which excels in real-time object detection, into this system. We trained the model for 20 epochs using a dataset of 49 Bangla signs, including letters and characters. A key innovation of this work is the addition of an avatar tutor to enhance BSL learning.

This system achieved a 95% detection rate for hand gestures in optimal lighting, but accuracy varied with environmental factors and gesture complexity. Future improvements will focus on refining accuracy across different conditions and incorporating user feedback for further enhancement.

Future research could focus on enhancing gesture recognition via AR and VR integration, expanding the BSL curriculum to include word and sentence recognition, and improving learning outcomes. Dynamic, customizable lesson plans and dedicated mobile and web applications for the avatar tutor could also boost accessibility and impact.

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